Gait asymmetry in transfemoral amputees across walking speeds

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1. Introduction

Individuals with a lower-limb amputation tend to rely more on their intact limb during gait. Over the years, this asymmetric loading has been associated with secondary conditions such as osteoarthritis of the intact knee and hip and low back pain (Gailey et al., 2008; Struyf et al., 2009).

Physical activity and sport are important for the health of people with an amputation, but they involve faster movements and higher loads than slow walking. If the intact limb compensates even more at higher speeds, more intense activities could further increase its overload.

For above-knee (transfemoral) amputees the problem is particularly relevant, because both the knee and the ankle are replaced by passive or semi-active prosthetic components, which cannot absorb the impact at heel strike or provide active push-off as the biological joints do.

Aim. The aim of this short project was to investigate whether the asymmetry between the intact and the prosthetic limb increases with walking speed, using an open biomechanics dataset of transfemoral amputees.

2. Data and methods

Dataset. I used the open dataset by Hood et al. (2020), which includes 18 individuals with a unilateral above-knee amputation walking on an instrumented treadmill at five speeds. Subjects were divided into two groups based on mobility level: K2 (0.4–0.8 m/s) and K3 (0.6–1.4 m/s), with nine subjects each. The dataset provides hip, knee and ankle angles (deg) and moments (Nm/kg) for the intact (contralateral) and prosthetic (ipsilateral) side, time-normalised to the gait cycle (1001 samples) and averaged across strides. Data were low-pass filtered (Butterworth, 6 Hz for markers and 15 Hz for forces) and processed in Visual3D by the dataset authors. This analysis focuses on the K3 group, whose speed range is closer to more active, faster movements.

Parameters. From the mean curves of each subject and speed, I extracted four peak values that are clearly defined on both sides:

Parameter	Window (% of gait cycle)	Value
Hip moment, early stance	0–20%	absolute value of the minimum
Hip moment, late stance	30–70%	maximum
Ankle moment, push-off	30–70%	absolute value of the minimum
Knee flexion, swing	50–90%	absolute value of the minimum angle

In addition, I extracted the peak knee extensor moment of the intact limb (0–30% of the cycle). This parameter has no equivalent on the prosthetic side, where the knee stays extended during stance, so it was analysed on the intact side only.

Symmetry index. For each parameter, asymmetry was quantified with the symmetry index:

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SI = \frac{X_{intact} - X_{prosthetic}}{0.5\,(X_{intact} + X_{prosthetic})} \times 100
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where SI = 0% indicates perfect symmetry, a positive SI a higher value on the intact side and a negative SI a higher value on the prosthetic side.

Statistics. For each K3 subject I fitted a linear regression of each parameter (SI, or the intact knee moment) against walking speed, and tested whether the mean of the nine slopes differed from zero with a one-sample t-test. Because five tests were performed, a Bonferroni-corrected threshold of $p < 0.01$ was used. The analysis was performed in Python (NumPy, pandas, SciPy, Matplotlib); code and figures are available at github.com/Martina-Agostini/gait-asymmetry-amputees.

K2 vs K3. The same slope analysis was repeated for the K2 group (0.4–0.8 m/s). Because the two groups walked at different speed ranges, their slopes cannot be compared directly; the groups were therefore compared at the two common speeds (0.6 and 0.8 m/s) with Welch's t-test, using a Bonferroni-corrected threshold of $p < 0.005$ for the 10 comparisons.

3. Results

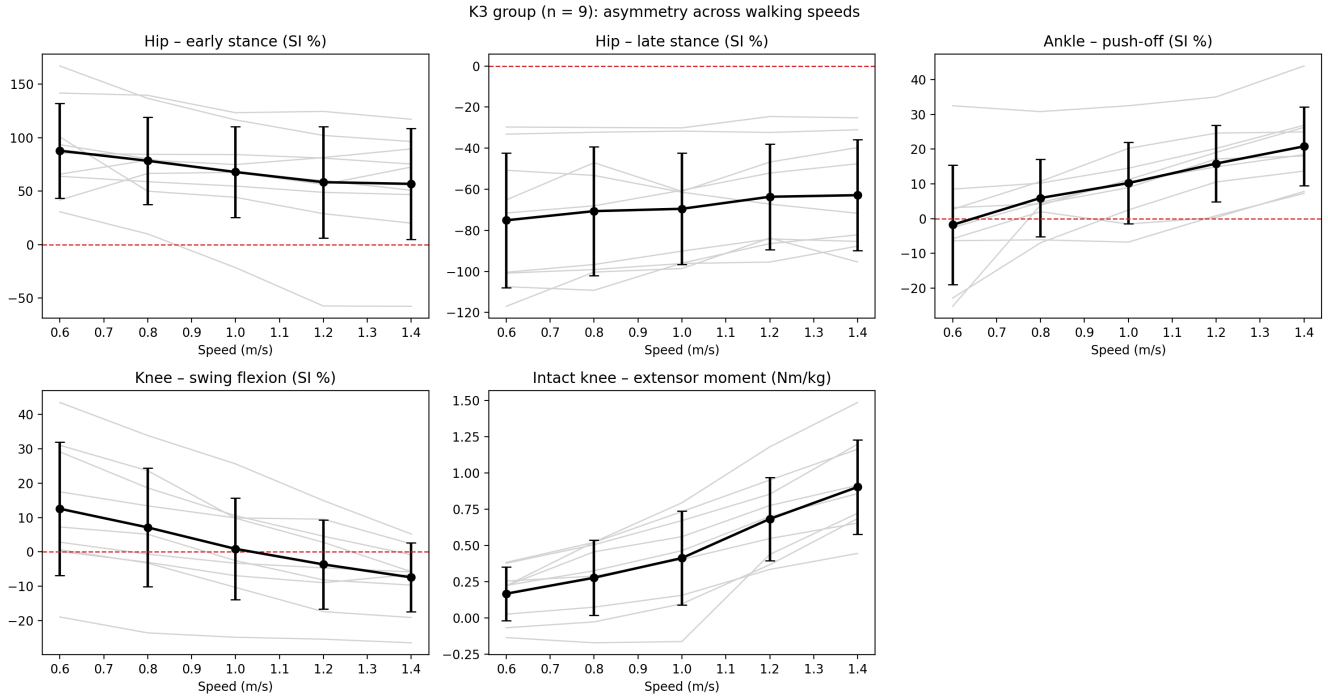


Figure 1. Symmetry index (SI) and intact knee extensor moment as a function of walking speed in the K3 group (n = 9). Grey lines: individual subjects; black: group mean ± SD; red dashed line: perfect symmetry.

Parameter	Mean slope (per m/s)	Subjects increasing	p-value
Intact knee extensor moment	+0.94 Nm/kg	9/9	< 0.001
Ankle push-off (SI)	+27.6 points	9/9	0.0007
Knee swing flexion (SI)	−25.4 points	0/9	0.0012
Hip late stance (SI)	+15.7 points	8/9	0.046
Hip early stance (SI)	−41.0 points	2/9	0.045

Table 1. Per-subject regression slopes against speed (K3 group) and one-sample t-test. Bonferroni threshold: p < 0.01.

Intact knee. The peak knee extensor moment of the intact limb increased from about 0.17 Nm/kg at 0.6 m/s to about 0.90 Nm/kg at 1.4 m/s. This peak reflects the quadriceps controlling knee flexion at impact. On the prosthetic side it is absent, because the prosthetic knee remains extended during stance, so shock absorption relies entirely on the intact limb.

Ankle push-off. At 0.6 m/s the two sides were similar (SI about −2%), while at 1.4 m/s the SI reached about +21%. The intact ankle increased its push-off moment with speed, while the prosthetic foot, a passive spring with fixed stiffness, changed much less. The additional push-off needed to walk faster was therefore provided mainly by the intact limb.

Knee swing flexion. The SI decreased from about +12% to about −7%. At higher speeds the prosthetic knee flexed as much as, or more than, the intact knee, probably because the prosthetic knee control reduces swing resistance at faster walking speeds.

Hip. In late stance the prosthetic side showed a higher peak at all speeds (SI between about −75% and −63%). This reflects a compensation: the hip on the prosthetic side has to pull the prosthesis forward into swing, because the prosthesis provides no active push-off. Neither hip parameter reached significance after Bonferroni correction.

K2 vs K3. In the K2 group, only knee swing flexion changed significantly with speed (SI slope −55.6 points per m/s, 1/9 subjects increasing, p = 0.007), in the same direction as in the K3 group. The intact knee extensor moment stayed almost constant at about 0.15 Nm/kg (p = 0.06), and the ankle push-off asymmetry remained at about +17% at all speeds (p = 0.93). For hip late stance the SI was highly variable (group SD of several hundred points), because in some subjects the peak was close to zero, so this parameter was not interpreted for the K2 group.

At the common speeds of 0.6 and 0.8 m/s the two groups were similar. The only difference with p < 0.05 was the ankle push-off SI at 0.6 m/s (K2 +17% vs K3 −2%, p = 0.048), which was not significant after Bonferroni correction.

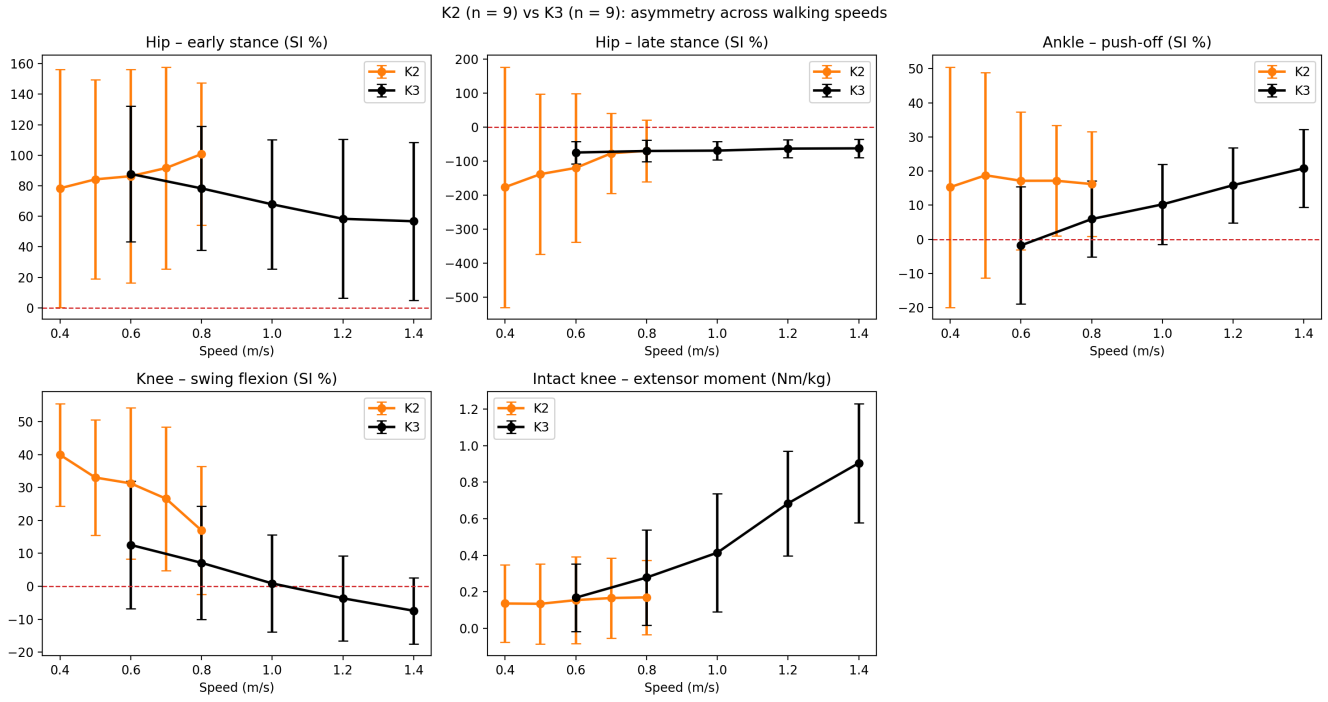


Figure 2. Symmetry index and intact knee extensor moment across walking speeds in the K2 (orange, n = 9) and K3 (black, n = 9) groups. Mean ± SD; red dashed line: perfect symmetry.

4. Discussion and limitations

In the K3 group, the load-related parameters showed that the intact limb takes on an increasing share of the load as walking speed increases: the intact knee bears a growing extensor moment, and the intact ankle provides most of the additional push-off. In contrast, the prosthesis adapted in movement (knee flexion in swing), but not in force production.

This pattern is consistent with the passive nature of the prosthetic components. A prosthetic foot with fixed stiffness cannot increase its push-off on demand, and a knee that remains extended during stance cannot absorb the impact, so the intact limb and the prosthetic-side hip have to compensate.

The comparison with the K2 group supports the role of walking speed. At the same speeds the two groups were similar, and in the K2 group, which walked only up to 0.8 m/s, the load on the intact limb did not clearly increase. The increasing overload observed in the K3 group therefore seems to be related mainly to the higher speeds reached, rather than to the mobility level itself. Swing knee flexion was the only parameter that changed consistently with speed in both groups.

These findings refer to walking up to 1.4 m/s. Since sport involves even faster movements and higher loads, the overload of the intact limb may be further amplified, which could contribute to the long-term joint problems described in amputees. This hypothesis, however, needs to be tested directly.

Limitations.

- No control group: the intact knee extensor moment also increases with speed in non-amputees, so it is not possible to say whether it increases *more* in amputees.
- Walking only, up to 1.4 m/s: running and sport movements were not analysed.
- Small samples (n = 9 per group) and above-knee amputees only. The K2 group covered a narrower speed range (0.4 m/s vs 0.8 m/s for K3), so its slopes are less precise, and K2 subjects may have relied more on the handrails.
- Peak moments were analysed, not mechanical work or cumulative load over time.
- The symmetry index is unstable when values are small (e.g. hip early stance at low speed, and hip late stance in the K2 group).

- Prosthetic-side moments are less reliable, because the prosthesis inertial properties were approximated and the hip joint centre was affected by movement at the socket-tissue interface (Hood et al., 2020).
- Some subjects used the treadmill handrails, which were not instrumented. Handrail use was not taken into account in this analysis.
- The data were collected and processed by the dataset authors; this project is a secondary analysis.

5. Next steps

This analysis could be extended in several directions:

- comparing the results with a group of non-amputee subjects walking at the same speeds;
- extending the analysis to running and sport-specific movements;
- including below-knee amputees and different prosthetic components;
- using musculoskeletal modelling to estimate internal joint loads, not only net joint moments.

References

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